From hypothesis space → therapeutic design space

Building a world model to explore and optimize therapeutic design space

An AI system that designs an AAV capsid for a gene therapy — navigating efficacy, immune escape, and safety in a space of more than 10^{11} possibilities.

LAYER 1
Generative

LAYER 2
Feature engineering

LAYER 3
World model (GP)

LAYER 4
RL closed loop

LAST SPRINT

An AI system generated **six drug targets** for age-related macular degeneration in **four minutes**.

One was complement gene **C9** — a variant driving excess complement activation in the back of the eye, destroying photoreceptors.

We have the target.
How do we design the therapeutic?

The design space exceeds 10^{11}. Directed evolution samples it blind.

735
amino acids in the AAV2 capsid shell

~60
residues control cell entry *and* immune escape

50–70%
adult seroprevalence — pre-existing antibodies

>10^{11}
engineerable variants in the design space

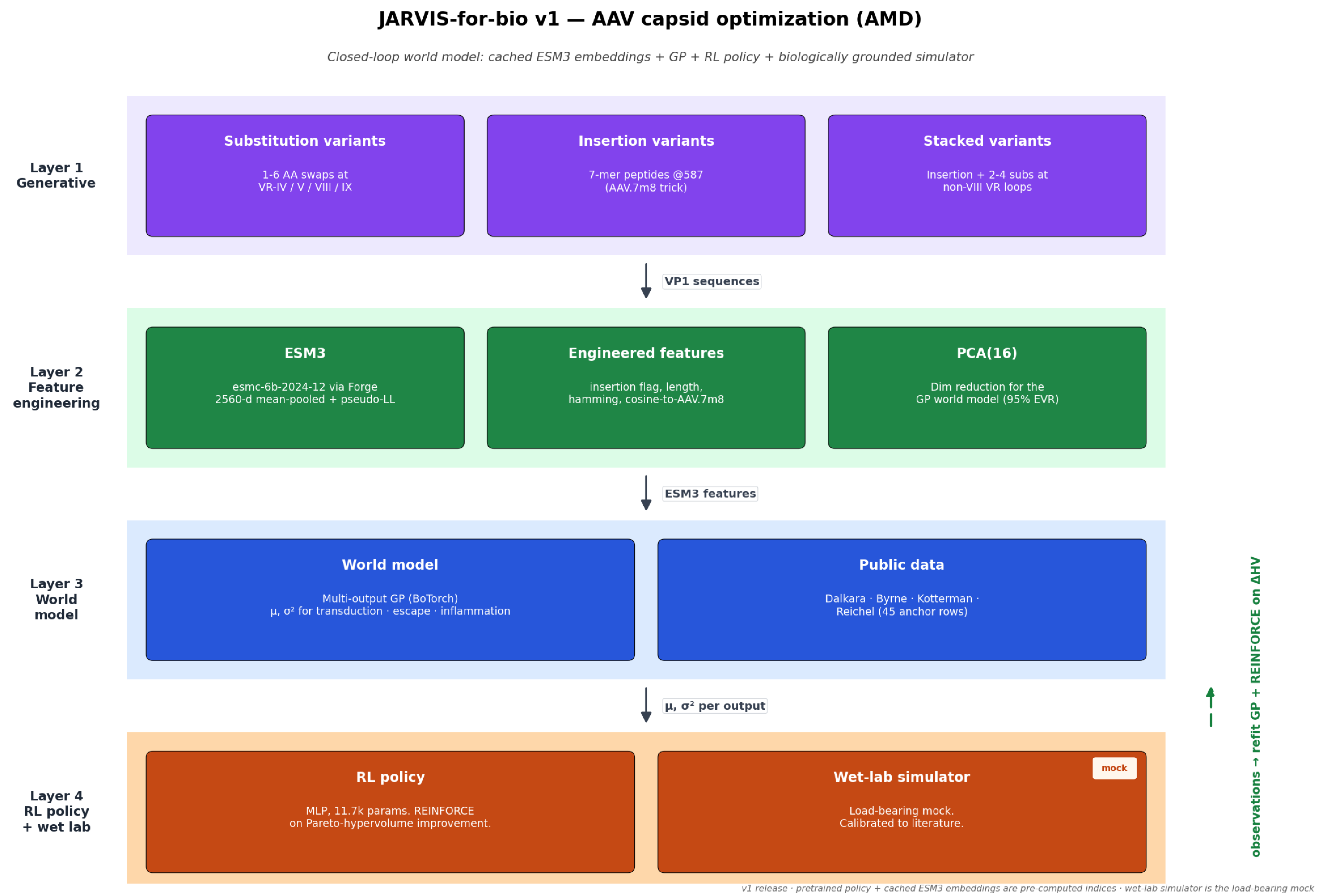
Gene therapy delivers a gene to retinal cells via an engineered virus — **AAV2**. Directed evolution works, but it's **blind sampling**: no model of the space, no way to navigate tradeoffs, no principled way to enforce a safety constraint during selection.

A world model changes that.

HOW IT WORKS · 1:20

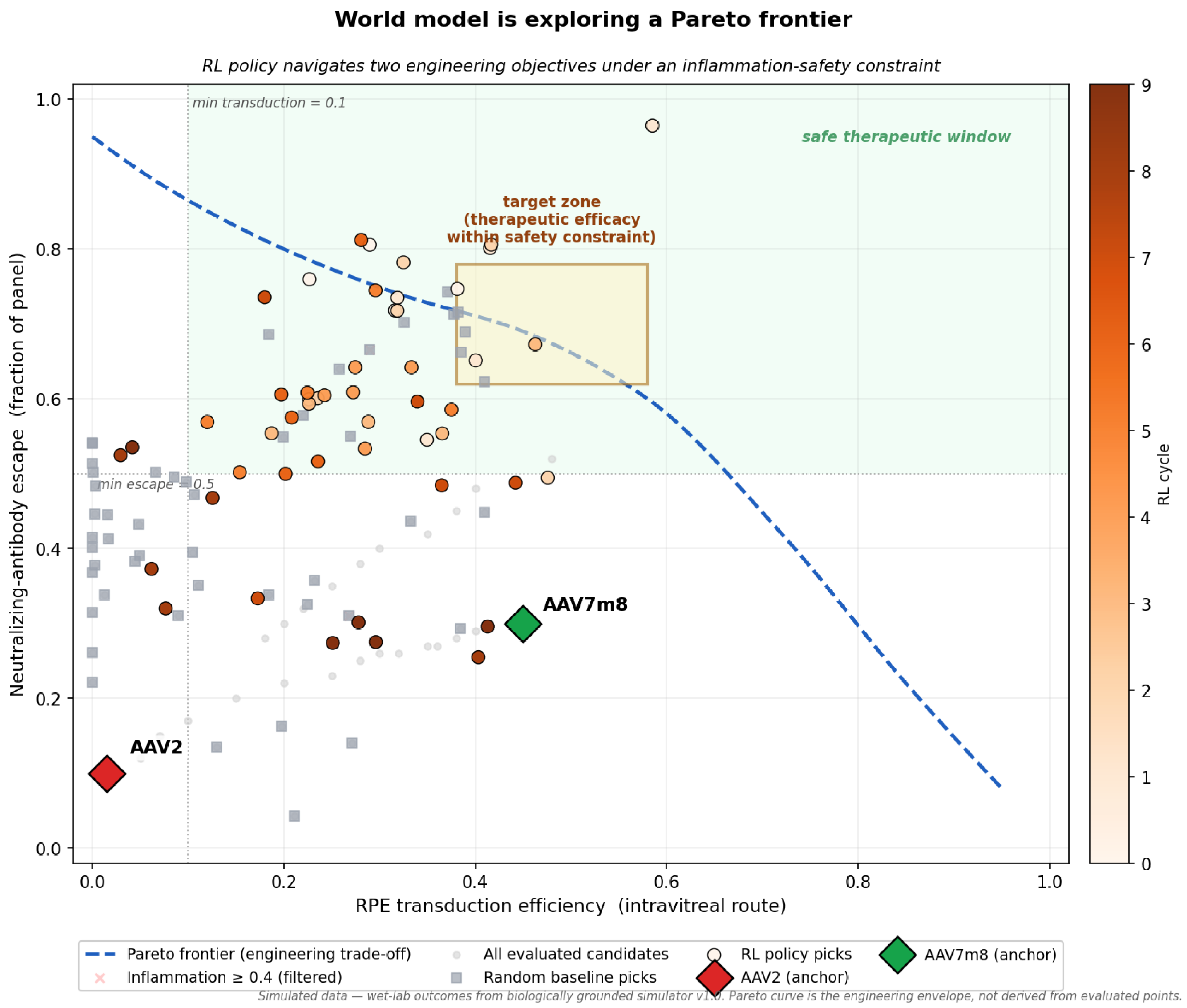
Four layers, one closed loop

Generate ~80 variants
ESM3 → 2560-d
Multi-output GP + σ²
RL policy refits GP



Optimizing three competing objectives

- **AAV2 wildtype** — the baseline. Low transduction, low escape.
- **AAV.7m8** — directed-evolution state of the art.
- **Pareto frontier** — where you can't improve one objective without sacrificing the other.
- **RL policy picks** climb into the target zone — efficacy within the safety constraint.

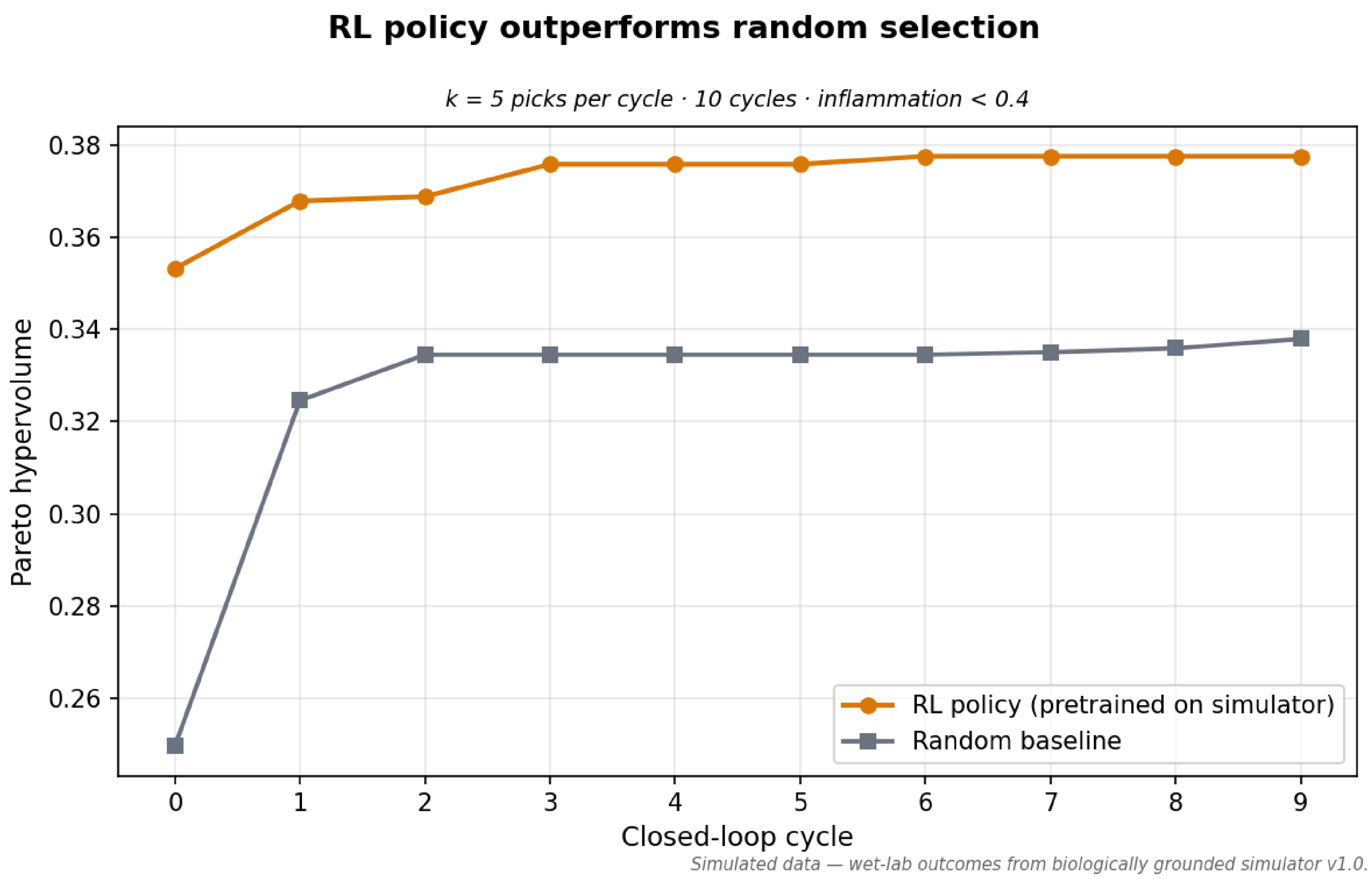


RL Policy beats random

Each cycle, observations feed back, the GP refits, and the policy updates. It reaches a higher Pareto hypervolume than random selection and keeps pulling away.

RL POLICY **~0.38**

RANDOM **~0.34**



Last sprint, AI explored hypothesis space. This sprint, AI explored therapeutic design space.

Same architecture. One leg further.

The wet-lab simulator is the only synthetic component — calibrated to published intravitreal AAV literature. Everything else is real. When real wet-lab results arrive, you swap them in; the policy interface doesn't change.

Full write-up, figures & code
→ in the repo